

Manuscript: "How Many People Live Near Protected Areas in Developing Countries? Estimates from Gridded Population Data (2000–2020)"

We thank the referees for their constructive comments. Below we respond to each point. Referee comments are in italics and our responses follow. Changes in the manuscript are referenced by section.

Important note: correction of a computation issue and reimplementation in R

While preparing the additional decomposition requested by Referee 3, we identified a technical issue in the original (Google Earth Engine, GEE) computation that was not raised by the reviewers but that we considered necessary to correct before resubmission. We flagged this to the editorial office and were granted a short extension.

The issue had two components:

1. Double counting of overlapping buffer zones: In the GEE workflow, the 2020 "all PAs" cross-section was assembled by summing two independently generated layers (PAs with a recorded designation year and PAs with a missing designation year). Where the 10 km buffer rings of these two layers overlapped, population and area were counted twice, inflating the "inside or within 10 km" figures.

2. Cross-border attribution: In the way that PAs were filtered, PAs located in neighbouring countries could contribute population to a country's totals where buffers crossed borders. This was not intentional, as clarified after the first review.

Google recently limited access to GEE for free research purposes, so we reimplemented the entire population-extraction pipeline in R, which also improves reproducibility. The corrected pipeline (i) computes the 2020 all-PA cross-section directly from a single joint mask, with mutually exclusive PA interiors and buffer rings and an exclusive category hierarchy (strict/non-strict/unknown), and (ii) filters PAs to each country by ISO3 code.

Effect on results: Aggregate magnitudes and all qualitative conclusions are unchanged, but the "inside or within 10 km" figures are lower than in the first-revision GEE version, mainly because of the corrected double counting. The population inside PAs is essentially unchanged. Specifically, for 2020 across the 75 LMICs (excluding India):

- Inside PAs: 60.8 million (2.5%) (previously ~61 million, 2.5%).
- Inside or within 10 km: 732 million (29.5%) (previously ~902 million, 36.6%).
- 2000 (PAs with recorded designation year): 19 million (1.1%) inside and 258 million (15.4%) inside or within 10 km.

The Methods (main text) and Supplementary Materials Section A now describe the R workflow and this correction; the replication package on GitHub/Software Heritage includes both the R code and the superseded GEE code.

Referee 1

The authors have performed a useful analysis of an important question... I have no further comments.

We thank Referee 1 for the positive assessment.

Referee 3

I would like to see a bit more nuance however - for example, it is not clear whether there are more people living within close proximity to non-strict PAs because there are more non-strict PAs to live close too (and/or within) or because of some other settlement pattern related to the PA type. This would be quite straightforward to address by detailing the number and coverage of each type of PA and then determining a standardized metric to compare (e.g. people within 10 km proximity / sq km of strict PA vs people within 10 km proximity / sq km of non-strict PA).

We now address this directly with an area-standardized decomposition (new paragraph in Results, "Population magnitudes", new standardized table in the Supplementary Materials). After dividing nearby population by PA area, non-strict PAs remain somewhat more populated than strict PAs (149 vs 118 per km² people inside or within 10 km of PA), but the larger raw population near non-strict PAs is driven mainly by scale: non-strict PAs are both more numerous (2,564 vs 1,160 sites) and slightly more extensive (1.39 vs 1.35 million km²). We now state that the raw difference reflects primarily PA number and area, with only a modest additional contribution from higher nearby population intensity.

It also might be worth detailing a bit more what the actual differences are between the two PA types - briefly - instead of just pointing to the IUCN categories, as these differences are likely to impact settlement practices.

We added a short characterization in the Discussion ("PA categories matter for interpreting impacts"): strict categories (IUCN I-III) are typically designated as areas where human habitation and extraction are prohibited or tightly limited, whereas non-strict categories (IUCN IV–VI) explicitly accommodate settlement and sustainable use.

Abstract – Ln 9: It would be nice to have the total global population that these 75 LMICs represent as this would quantify the scale of the analysis.

The abstract now states the combined population of the 75 LMICs (about 2.48 billion people in 2020).

Abstract – Ln 11: I initially thought there was no need to explicitly state that India is not included in the abstract unless it is the ONLY LMIC not included – which it is not, especially since India is correctly explicitly mentioned in the methods as its population size etc. make it a sizeable omission. However, given its outsized influence, on reflection it is probably worth keeping this in the abstract, however I think you should then provide a simple justification such as "India, representing 36% of LMIC's global population, is excluded..." This obviously points out a weakness in the manuscript right off the bat but is a proper and honest approach.

The abstract now reads "India, which represents about 36% of the LMIC population, is excluded because its national protected areas are not publicly available in the WDPA."

Abstract – Ln 14: Show the numbers and percentages here for 2000 so the difference can be appreciated.

The abstract now reports the 2000 figures alongside 2020: 19 million (1.1%) inside and 258 million (15.4%) inside or within 10 km.

Pg 1, Ln 36 – Pg 2, Ln 13: This entire paragraph should be in the Methods section as it is a justification for the process. In the Introduction it is enough to detail the concept of those living in proximity to PAs being affected by PA policy etc. The nitty-gritty details of the exact proximity is a methodological question. What you say in Ln 19-20 (“...within a reference buffer commonly used in the literature...”) is sufficient for the Intro.

The exact proximity definitions are now given in Methods ("Study Area"), and the Introduction retains only the concept, stating "within a reference buffer commonly used in the literature".

Pg 4 Ln 11: What % of the pop did this 17.9 million people represent?

The corresponding percentage is now reported throughout (inside PAs in 2000 corresponds to 1.1% of the 2000 sample population; the 2020 inside-PA figure of 60.8 million corresponds to 2.5%).

Table 1: The obvious omission is why the numbers “inside Strict PAs only”, “inside Non-strict PAs only” and “inside Unknown category PAs only” are not included here?

Table 1 has been extended to include inside-only rows by category. For 2020, of the 60.8 million people inside PAs, 10.4 million are inside strict PAs, 19.0 million inside non-strict PAs, and 31.5 million inside PAs of unrecorded IUCN category; the corresponding inside-or-within-10 km rows are 159 / 207 / 366 million.

Pg 9 Ln 13-15: Is this because there are fewer strict protection regimes, or because there is some fundamental difference in settlement patterns? I suspect the former. It is worth teasing this apart. Is it not also the case that many strict protection PAs do not allow populations to live within their boundaries?

We now mention these apart. The standardization shows the difference is mainly one of scale (number and area of PAs) rather than of nearby population intensity. We also note explicitly that the near-absence of residents inside strict PAs is partly by design, since strict categories commonly prohibit human habitation.

Pg 9 Ln 16-20: This functional difference in PA type has been explicitly shown in ecological studies of wildlife. For example, habitat suitability for leopards in Sri Lanka is strongly improved by close proximity to “Level 1 PAs” (analogous to “strict protection regimes”) but unaffected by “Level 2 PAs” (analogous to “non-strict PAs” where entrance and extraction are allowed) - Kittle, A.M., Watson, A.C., Cushman, S.A. & Macdonald, D.W. 2018. Forest cover and level of protection influence the island-wide distribution of an apex carnivore and umbrella species, the Sri Lankan leopard (Panthera pardus kotiya). Biodiversity and Conservation 27(1): 235-263. <https://doi.org/10.1007/s10531-017-1431-8>

We added this result and citation to the Discussion: habitat suitability for the Sri Lankan leopard is improved by proximity to strictly protected areas but unaffected by less strictly protected areas where entry and extraction are permitted (Kittle et al. 2018), illustrating that management category differences are functionally meaningful beyond human populations.

Conclusion

We believe these revisions, together with the corrected and now fully reproducible R pipeline, substantially strengthen the manuscript, and we thank the referees again for their input.